

Paper 1 - Set 02 - Solutions

Generated practice paper: answer key and worked solutions.

Answer key

Question	Answer	Topic
1	C	Simultaneous equations and inequalities
2	A	Angles, polygons, congruence and vectors
3	F	GCSE algebra
4	C	Exponentials and logarithms
5	A	Functions and transformations
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9	C	Triangles, radians and sectors
10	E	Probability
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19	E	Quadratic functions
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Worked solutions

Question 1 SIM-09 · Simultaneous equations and inequalities · spec MM1.4

The simultaneous equations $y = x^2 + 2x + c$ and $y = 4x + 1$ have exactly one solution. Find the value of c .

Answer: C

$x^2 - 2x + (c - 1) = 0$ has one solution when $4 - 4(c - 1) = 0$, so $c = 2$.

Full method - how and why

'Exactly one solution' for a line and a parabola is again the discriminant-zero (tangency) condition.

- 1 Set the expressions equal: $x^2 + 2x + c = 4x + 1$.

why: A simultaneous solution is a point where both y -values agree.

- 2 Collect: $x^2 - 2x + (c - 1) = 0$.

why: Move everything to one side; the $2x - 4x$ gives $-2x$.

- 3 Discriminant zero: $(-2)^2 - 4(1)(c - 1) = 0$.

why: One solution means a repeated root.

- 4 Solve: $4 - 4c + 4 = 0 \Rightarrow 4c = 8 \Rightarrow c = 2$.

why: Careful with $-4(c - 1) = -4c + 4$.

Where it goes wrong

- Mishandling $-4(c - 1)$ and getting $c = 0$.
- Setting the discriminant > 0 ('has a solution') rather than $= 0$ ('exactly one').

Takeaway Exactly one intersection between a line and a parabola always means discriminant $= 0$.

Question 2 GEO-08 · Angles, polygons, congruence and vectors · spec M5.4

Which of the following sets of information is NOT sufficient to prove that two triangles are congruent?

Answer: A

SSA is not a congruence condition (it is the ambiguous case of the sine rule); the other four are the standard criteria.

Full method - how and why

The four congruence criteria are SSS, SAS, ASA and RHS. Anything else - notably SSA - is not sufficient.

- 1 Recall the valid criteria: SSS, SAS, ASA (or AAS), RHS.

why: These four are the ones that fix a triangle completely.

- 2 Consider SSA: two sides and a **non-included** angle. This is the ambiguous case of the sine rule, where two different triangles can share the same data.

why: The side opposite the known angle can 'swing' to two positions - exactly the situation in TRI-07.

- 3 So SSA is the odd one out.

why: RHS is really a special case of SSA that works, because the right angle removes the ambiguity.

Where it goes wrong

- Thinking RHS is invalid because it looks like SSA - the right angle makes it unambiguous.
- Confusing AAA (similar but not congruent) with ASA.

Takeaway SSS, SAS, ASA, RHS prove congruence; SSA does not, and AAA gives only similarity.

Question 3 ALG-12 · GCSE algebra · spec M4.19

A sequence has first term 2 and each subsequent term is obtained by squaring the previous term and subtracting 2. What is the 50th term?

Answer: F

$$2^2 - 2 = 2: \text{ every term is } 2.$$

Full method - how and why

A recurrence with a fixed point. Computing one term reveals the sequence never changes.

- 1** Apply the rule to the first term: $u_2 = 2^2 - 2 = 2$.

why: 'Square then subtract 2' applied to 2 returns 2.

- 2** Since $u_2 = u_1$, the same calculation repeats forever: every term is 2.

why: 2 is a fixed point of the map $x \mapsto x^2 - 2$.

- 3** So $u_{50} = 2$.

why: No need to iterate fifty times.

Where it goes wrong

- Assuming the terms must grow because of the squaring, and choosing 2^{50} .
- Computing $2^2 - 2$ as 0 or -2 .

Takeaway Always compute the second term: if it repeats the first, the sequence is constant.

Question 4 LOG-11 · Exponentials and logarithms · spec MM5.2

Solve $\log_2 x + \log_4 x = 6$. (You may use $\log_4 x = \frac{1}{2}\log_2 x$.)

Answer: C

$$\frac{3}{2}\log_2 x = 6 \Rightarrow \log_2 x = 4 \Rightarrow x = 16.$$

Full method - how and why

Two different bases. The hint converts one to the other, after which it is a linear equation in $\log_2 x$.

- 1** Use the given relation $\log_4 x = \frac{1}{2}\log_2 x$.

why: It holds because $4 = 2^2$: it takes twice as many factors of 2 as of 4 to build x .

- 2** Substitute: $\log_2 x + \frac{1}{2}\log_2 x = 6$, i.e. $\frac{3}{2}\log_2 x = 6$.

why: Both terms now involve the same quantity, so treat $\log_2 x$ as a single unknown.

- 3** Solve for the log: $\log_2 x = 4$.

why: Multiply by $\frac{2}{3}$.

- 4** Convert: $x = 2^4 = 16$.

why: $\log_2 x = 4$ means $x = 2^4$.

Where it goes wrong

- Solving $\log_2 x = 6$ and answering 64 - which is offered (twice, as 64 and 2^6).
- Adding the bases or the arguments instead of using the relation.

Takeaway Get every logarithm to the same base first; then it is ordinary algebra in that one quantity.

Question 5 FUN-13 · Functions and transformations · spec MM8.4

The graph of $y = a(x + b)^2 + c$ has vertex $(-2, 5)$ and passes through $(0, -3)$. Find a .

Answer: A

$b = 2, c = 5$. At $x = 0$: $4a + 5 = -3$, so $a = -2$.

Full method - how and why

The completed-square form is handed to you, so the vertex gives two constants and the extra point gives the third.

- 1 Match the vertex: $y = a(x + b)^2 + c$ has vertex $(-b, c)$, and we are told it is $(-2, 5)$, so $b = 2$ and $c = 5$.

why: Note the sign: the form uses $x + b$, so the vertex x -coordinate is $-b$, not b .

- 2 Substitute the point $(0, -3)$: $a(0 + 2)^2 + 5 = -3$.

why: One point is enough to find the remaining unknown.

- 3 Solve: $4a = -8$, so $a = -2$.

why: Negative, which makes sense: the curve passes below its vertex, so it opens downwards.

Where it goes wrong

- Taking $b = -2$ and getting $a = -\frac{1}{2}$ from $a(0 - 2)^2 \dots$ (the same here, but the sign confusion bites when the point is not symmetric).
- Answering $+2$ and ignoring that the parabola must open downwards to reach $y = -3$.

Takeaway Sanity-check the sign of a against the picture: if the curve drops below the vertex, $a < 0$.

Question 6 TRG-12 · Trigonometric functions and equations · spec MM4.6

Find the number of solutions of $\tan^2 x = 3$ for $0 \leq x < 2\pi$.

Answer: B

$\tan x = \pm\sqrt{3}$: $x = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{4\pi}{3}, \frac{5\pi}{3}$. Four solutions.

Full method - how and why

A squared tangent equation. Two values of $\tan x$, each giving solutions once per π .

- 1 Take square roots: $\tan x = \pm\sqrt{3}$.

why: Both signs are needed.

- 2 $\tan x = \sqrt{3}$ gives $x = \frac{\pi}{3}$ and $\frac{\pi}{3} + \pi = \frac{4\pi}{3}$.

why: Period π for $\tan$.

- 3 $\tan x = -\sqrt{3}$ gives $x = \frac{2\pi}{3}$ and $\frac{2\pi}{3} + \pi = \frac{5\pi}{3}$.

why: $\tan$ is negative in quadrants 2 and 4; the reference angle is still $\frac{\pi}{3}$.

- 4 Four solutions in $[0, 2\pi)$.

why: Two values $\times$ two periods.

Where it goes wrong

- Taking only the positive root and answering 2.
- Using period 2π for $\tan$.

Takeaway $\tan^2 x = k$ gives $\tan x = \pm\sqrt{k}$, and each has one solution per interval of length π .

Question 7 DIF-03 · Differentiation · spec MM6.3

The tangent to $y = x^2 - 4x + 1$ at the point where $x = 3$ meets the y -axis at

Answer: F

$y' = 2x - 4 = 2$ at $x = 3$; the point is $(3, -2)$. Tangent: $y + 2 = 2(x - 3)$, so at $x = 0$, $y = -8$.

Full method - how and why

Tangent line: gradient from the derivative, point from the curve, then find where it meets the axis.

- 1 Differentiate: $y' = 2x - 4$; at $x = 3$ the gradient is 2.
why: The derivative evaluated at a point gives the tangent's gradient there.
- 2 Find the point: $y(3) = 9 - 12 + 1 = -2$, so the tangent touches at $(3, -2)$.
why: The tangent passes through the curve at that x - use the original function.
- 3 Write the tangent: $y + 2 = 2(x - 3)$.
why: Point-gradient form with the point and gradient just found.
- 4 Set $x = 0$: $y + 2 = -6$, so $y = -8$. The tangent meets the y -axis at $(0, -8)$.
why: The y -axis is $x = 0$.

Where it goes wrong

- Using $y(3) = -2$ as the intercept and answering $(0, -2)$.
- Sign slip: $y = 2x - 8$, not $2x + 8$.

Takeaway Tangent = gradient from f' + point from f ; then substitute $x = 0$ for the intercept.

Question 8 QUA-21 · Quadratic functions · spec MM1.3

The parabola $y = x^2 - 4x + 3$ is reflected in the x -axis and then translated 2 units in the positive y -direction. What is the maximum value of the resulting function?

Answer: A

$x^2 - 4x + 3 = (x - 2)^2 - 1$, minimum -1 . Reflecting gives maximum 1; translating up by 2 gives maximum 3.

Full method - how and why

Transformations applied to a quadratic. Track what happens to the **vertex** rather than to the whole equation.

- 1 Complete the square: $x^2 - 4x + 3 = (x - 2)^2 - 1$, so the minimum is -1 at $x = 2$.
why: The turning point is the only feature the transformations act on in a way that matters.
- 2 Reflect in the x -axis: $y \rightarrow -y$, so the minimum -1 becomes a **maximum** of 1.
why: Reflection in the x -axis flips the sign of every y -value, turning a minimum into a maximum.
- 3 Translate 2 up: the maximum becomes $1 + 2 = 3$.
why: A vertical translation adds to every y -value, including the extreme one.

Where it goes wrong

- Forgetting that the reflection converts the minimum into a maximum and reporting $-1 + 2 = 1$.
- Reflecting the x -coordinate as well - the turning point stays at $x = 2$ throughout.

Takeaway For transformation questions, follow one key point (usually the vertex) through each step in order.

Question 9 TRI-13 · Triangles, radians and sectors · spec MM4.2

The minute hand of a clock is 12 cm long. How far does its tip travel in 25 minutes?

Answer: C

25 minutes is $\frac{25}{60} = \frac{5}{12}$ of a revolution: $\frac{5}{12} \times 2\pi \times 12 = 10\pi$ cm.

Full method - how and why

Distance travelled by a rotating tip is arc length. Find the fraction of a full turn, then multiply by the circumference.

- Find the fraction of a revolution: $\frac{25}{60} = \frac{5}{12}$.
why: The minute hand takes 60 minutes for one full turn.
- Compute the full circumference: $2\pi r = 2\pi(12) = 24\pi$ cm.
why: The tip traces a circle of radius equal to the hand's length.
- Multiply: $\frac{5}{12} \times 24\pi = 10\pi$ cm.
why: Equivalently, $\theta = \frac{5}{12} \times 2\pi = \frac{5\pi}{6}$ and $s = r\theta = 12 \times \frac{5\pi}{6} = 10\pi$.

Where it goes wrong

- Using the radius as the distance, or the area formula instead of arc length.
- Using $\frac{25}{12}$ or $\frac{25}{60}$ of the **radius** rather than the circumference.

Takeaway Arc length = (fraction of a turn) $\times 2\pi r$; a clock hand makes one turn per hour.

Question 10 PRO-11 · Probability · spec M7.6

Two cards are drawn at random from a standard pack of 52 without replacement. Which of the following expressions is the probability that both are aces?

Answer: E

Without replacement the second draw has 3 aces among 51 cards.

Full method - how and why

The question asks for the **expression**, so the reasoning is entirely about how the second draw changes.

- First card an ace: $\frac{4}{52}$ - four aces in the pack.
why: All 52 cards are equally likely.
- Given that, one ace and one card overall have gone: 3 aces remain among 51 cards.
why: Without replacement, **both** numbers drop by one.
- Multiply: $\frac{4}{52} \times \frac{3}{51}$.
why: This is $P(A) \times P(B|A)$, the multiplication rule for dependent events.

Where it goes wrong

- $\frac{4}{52} \times \frac{4}{52}$ - that is with replacement.
- $\frac{4}{52} \times \frac{3}{52}$ - reducing the aces but forgetting the total also falls to 51.
- Adding rather than multiplying: 'and' means multiply.

Takeaway Without replacement, reduce numerator **and** denominator; 'and' $\Rightarrow$ multiply.

Question 11 NUM-15 · Number, units and estimation · spec M2.5

How many three-digit positive integers have digits that are all different and all odd?

Answer: F

Odd digits 1, 3, 5, 7, 9: $5 \times 4 \times 3 = 60$.

Full method - how and why

Counting with two restrictions: digits from a fixed set, and all different.

- 1 List the allowed digits: 1, 3, 5, 7, 9 - five odd digits.
why: 'All odd' restricts the pool; note 0 is even so no leading-zero problem arises.
- 2 First digit: 5 choices. Second: 4 (one used). Third: 3.
why: 'All different' removes one option at each stage.
- 3 Multiply: $5 \times 4 \times 3 = 60$.
why: The product rule.

Where it goes wrong

- Using $5^3 = 125$ by ignoring 'all different' - an offered option.
- Worrying about leading zeros, which cannot occur here.

Takeaway Count position by position, reducing the pool by one each time when repetition is banned.

Question 12 FUN-11 · Functions and transformations · spec MM8.2

The graph of $y = f(x)$ is transformed by the sequence: reflect in the x -axis, then translate 2 units in the positive y -direction, then reflect in the y -axis. The resulting graph has equation

Answer: D

Reflect in x -axis: $y = -f(x)$. Translate up 2: $y = -f(x) + 2$. Reflect in y -axis (replace x by $-x$): $y = 2 - f(-x)$.

Full method - how and why

Three transformations in a stated order. Apply them to the equation one at a time, and be careful that the last one acts on everything built so far.

- 1 Reflect in the x -axis: $y = -f(x)$.
why: Negating the output flips the graph vertically.
- 2 Translate 2 up: $y = -f(x) + 2$.
why: Add to the whole expression, not just to f .
- 3 Reflect in the y -axis: replace **every** x by $-x$, giving $y = -f(-x) + 2 = 2 - f(-x)$.
why: Only x is replaced; the constant 2 is unaffected because it does not involve x .

Where it goes wrong

- Writing $-f(-x) - 2$ by also negating the constant.
- Applying the transformations in the wrong order - with mixed inside/outside operations, order matters.

Takeaway Rewrite the equation after each step; a y -axis reflection substitutes $-x$ for x everywhere it appears.

Question 13 TRG-13 · Trigonometric functions and equations · spec MM4.5

For all θ where the expressions are defined, $(\sin \theta + \cos \theta)^2 + (\sin \theta - \cos \theta)^2$ equals

Answer: D

Expanding: $(1 + 2\sin \theta \cos \theta) + (1 - 2\sin \theta \cos \theta) = 2$.

Full method - how and why

Expanding two brackets whose cross terms cancel. The Pythagorean identity finishes it.

- 1 Expand the first: $(\sin \theta + \cos \theta)^2 = \sin^2 \theta + 2\sin \theta \cos \theta + \cos^2 \theta$.

why: Standard $(a + b)^2$ expansion.

- 2 Expand the second: $(\sin \theta - \cos \theta)^2 = \sin^2 \theta - 2\sin \theta \cos \theta + \cos^2 \theta$.

why: Only the middle sign changes.

- 3 Add: the $\pm 2\sin \theta \cos \theta$ terms cancel, leaving $2(\sin^2 \theta + \cos^2 \theta)$.

why: This cancellation is the point of the question.

- 4 Apply the identity: $2 \times 1 = 2$.

why: $\sin^2 + \cos^2 = 1$ for every θ , so the answer is a constant.

Where it goes wrong

- Keeping a cross term and choosing $2 + 2\sin \theta \cos \theta$ - an offered option.
- Answering 1 by forgetting the factor 2.

Takeaway $(a + b)^2 + (a - b)^2 = 2(a^2 + b^2)$; with sin and cos that is always 2.

Question 14 SIM-12 · Simultaneous equations and inequalities · spec MM1.4

The curves $y = x^2 - 4$ and $y = -x^2 + 2x + k$ intersect at exactly two points. Find the complete set of values of k .

Answer: C

$2x^2 - 2x - 4 - k = 0$; discriminant $4 + 8(4 + k) > 0 \Rightarrow 36 + 8k > 0 \Rightarrow k > -\frac{9}{2}$.

Full method - how and why

Two parabolas meeting twice. Subtracting gives one quadratic, and 'two distinct points' is a positive discriminant.

- 1 Set them equal: $x^2 - 4 = -x^2 + 2x + k$.

why: Intersections satisfy both equations, so their right-hand sides agree.

- 2 Collect everything on one side: $2x^2 - 2x - 4 - k = 0$.

why: Note the $-x^2$ becomes $+x^2$ when moved across, giving $2x^2$.

- 3 Discriminant > 0 : $(-2)^2 - 4(2)(-4 - k) > 0$.

why: Here $c = -4 - k$; keeping the whole thing in brackets avoids sign errors.

- 4 Simplify: $4 + 32 + 8k > 0 \Rightarrow 8k > -36 \Rightarrow k > -\frac{9}{2}$.

why: $-4 \times 2 \times (-4 - k) = 32 + 8k$ - two negatives make a positive.

Where it goes wrong

- Losing a sign in $c = -4 - k$ and getting $k > \frac{9}{2}$ or $k > -4$, both of which are offered.
- Forgetting to combine x^2 with $-(-x^2)$ and working with x^2 instead of $2x^2$.

Takeaway Bring both curves to one side before using the discriminant, and bracket the constant term to keep signs straight.

Question 15 SEQ-08 · Sequences and series · spec MM2.3

The second term of a geometric sequence is 6 and the fifth term is 48. Find the sum of the first 6 terms.

Answer: F

$$ar = 6, ar^4 = 48, \text{ so } r^3 = 8, r = 2, a = 3. S_6 = 3 \cdot \frac{2^6 - 1}{2 - 1} = 3 \times 63 = 189.$$

Full method - how and why

Two terms of a geometric sequence determine a and r . Dividing the two equations isolates r .

- 1 Write the facts: $ar = 6$ and $ar^4 = 48$.

why: The k th term is ar^{k-1} ; the second term has r^1 , the fifth has r^4 .

- 2 Divide: $\frac{ar^4}{ar} = r^3 = \frac{48}{6} = 8$, so $r = 2$.

why: Division cancels a - always the first move with two geometric terms.

- 3 Back-substitute: $a(2) = 6$, so $a = 3$.

why: Use the simpler of the two equations.

- 4 Sum: $S_6 = a \frac{r^6 - 1}{r - 1} = 3 \times \frac{64 - 1}{1} = 189$.

why: With $r = 2$ the denominator is 1, making the arithmetic easy.

Where it goes wrong

- Using $r^4/r^1 = r^3$ instead of r^3 and getting a non-integer ratio.
- Computing S_6 with $2^6 = 32$ instead of 64.

Takeaway Divide two geometric terms to find r ; the index difference tells you which power of r you get.

Question 16 POL-13 · Polynomials, factor and remainder theorems · spec MM1.6

Which of the following is a factor of $x^4 - 5x^2 + 4$?

Answer: E

$$x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4) = (x - 1)(x + 1)(x - 2)(x + 2).$$

Full method - how and why

A quartic that is quadratic in x^2 . Factorise fully and read off which of the offered factors appears.

- 1 Substitute $u = x^2$: $u^2 - 5u + 4 = (u - 1)(u - 4)$.

why: Only even powers appear, so the substitution works.

- 2 Undo it: $x^4 - 5x^2 + 4 = (x^2 - 1)(x^2 - 4)$.

why: Replace u by x^2 in each bracket.

- 3 Factorise each as a difference of two squares: $(x - 1)(x + 1)(x - 2)(x + 2)$.

why: $x^2 - a^2 = (x - a)(x + a)$ - do not stop at the quadratics.

- 4 Compare with the options: $(x + 2)$ appears.

why: Equivalently, a factor $(x - a)$ exists iff $f(a) = 0$: $f(-2) = 16 - 20 + 4 = 0$ ✓.

Where it goes wrong

- Choosing $x^2 + 1$ or $x^2 + 4$ - these are sums, not differences, and are not factors.
- Testing only positive values when checking $f(a) = 0$.

Takeaway Factorise as far as possible; a quick check of $f(a) = 0$ confirms any candidate factor $(x - a)$.

Question 17 INT-15 · Integration · spec MM7.1

The area between the curve $y = x^2$ and the line $y = k$ (where $k > 0$) is $\frac{32}{3}$. Find k .

Answer: D

$$\text{Area} = \int_{-\sqrt{k}}^{\sqrt{k}} (k - x^2) dx = 2 \left(k^{3/2} - \frac{k^{3/2}}{3} \right) = \frac{4}{3} k^{3/2} = \frac{32}{3} \Rightarrow k^{3/2} = 8 \Rightarrow k = 4.$$

Full method - how and why

Work backwards from a given area to the parameter. Symmetry halves the work.

- Find the intersections: $x^2 = k$ at $x = \pm\sqrt{k}$.
why: The horizontal line cuts the parabola at two symmetric points.
- Set up the area: $\int_{-\sqrt{k}}^{\sqrt{k}} (k - x^2) dx$, with the line above the curve between the roots.
why: Upper minus lower.
- Use symmetry: $= 2 \int_0^{\sqrt{k}} (k - x^2) dx = 2 \left[kx - \frac{x^3}{3} \right]_0^{\sqrt{k}} = 2 \left(k^{3/2} - \frac{k^{3/2}}{3} \right) = \frac{4}{3} k^{3/2}$.
why: The integrand is even, so integrate over half the range and double. Note $k\sqrt{k} = k^{3/2}$.
- Solve $\frac{4}{3} k^{3/2} = \frac{32}{3}$: $k^{3/2} = 8$, so $k = 8^{2/3} = 4$.
why: Undo the $\frac{3}{2}$ power by raising to the $\frac{2}{3}$: $8^{2/3} = (\sqrt[3]{8})^2 = 4$.

Where it goes wrong

- Solving $k^{3/2} = 8$ as $k = 8^{3/2} = 22.6$ or $k = \sqrt{8}$ - both offered in spirit.
- Forgetting to double after using symmetry.

Takeaway Use even symmetry to halve the integration; undo a $\frac{3}{2}$ power by raising to $\frac{2}{3}$.

Question 18 LOG-14 · Exponentials and logarithms · spec MM5.3

The number of integers n satisfying $2^n < 1000 < 2^{n+3}$ is

Answer: A

$$2^9 = 512 < 1000 < 1024 = 2^{10}. \text{ Need } n \leq 9 \text{ and } n+3 \geq 10, \text{ so } n = 7, 8, 9: \text{ three integers.}$$

Full method - how and why

A double inequality in powers of 2. Locate 1000 between consecutive powers, then translate into conditions on n .

- Find where 1000 sits: $2^9 = 512$ and $2^{10} = 1024$, so $2^9 < 1000 < 2^{10}$.
why: Powers of 2 up to 1024 are worth knowing by heart.
- The left condition $2^n < 1000$ requires $n \leq 9$.
why: $2^{10} = 1024 > 1000$, so $n = 10$ fails.
- The right condition $1000 < 2^{n+3}$ requires $n+3 \geq 10$, i.e. $n \geq 7$.
why: $2^9 = 512 < 1000$, so $n+3 = 9$ (i.e. $n = 6$) fails.
- Combine: $7 \leq n \leq 9$, giving $n = 7, 8, 9$ - three integers.
why: Count the integers in the range, inclusive.

Where it goes wrong

- Counting $9 - 7 = 2$ instead of 3.
- Using strict inequalities on n where the powers allow equality.

Takeaway Convert conditions on 2^n into conditions on n by comparing with the nearest powers, then count inclusively.

Question 19 QUA-09 · Quadratic functions · spec MM1.3

Find the sum of all real values of x satisfying $(x^2 - 3x)^2 - 2(x^2 - 3x) - 8 = 0$.

Answer: E

Let $u = x^2 - 3x$: $(u - 4)(u + 2) = 0$. $x^2 - 3x - 4 = 0$ gives $x = 4, -1$ (sum 3). $x^2 - 3x + 2 = 0$ gives $x = 1, 2$ (sum 3). Total 6.

Full method - how and why

A quadratic in the expression $x^2 - 3x$. After substituting back you get two quadratics, and the question only wants the total of all four roots.

- 1 Let $u = x^2 - 3x$; the equation becomes $u^2 - 2u - 8 = 0$.

why: The same expression appears squared and linearly, which is the signal for a substitution.

- 2 Factorise: $(u - 4)(u + 2) = 0$, so $u = 4$ or $u = -2$.

why: Both values are usable - neither is ruled out, because $x^2 - 3x$ can take any value $\geq -\frac{9}{4}$.

- 3 Case $u = 4$: $x^2 - 3x - 4 = 0$. Its roots sum to 3 (sum $= -b/a$). Discriminant $9 + 16 > 0$, so both are real.

why: You never need the roots themselves - the sum of roots comes straight from the coefficients.

- 4 Case $u = -2$: $x^2 - 3x + 2 = 0$. Roots sum to 3; discriminant $9 - 8 > 0$, so both are real.

why: Check the discriminant in each case: a case with no real roots would contribute nothing to the total.

- 5 Total: $3 + 3 = 6$.

why: Four real values of x , summing to 6.

Where it goes wrong

- Answering 3 by solving only one of the two cases.
- Adding the u values ($4 + (-2) = 2$) instead of the x values.

Takeaway With a substitution, each value of u gives its own equation in x ; check each for real roots and use 'sum of roots $= -b/a$ ' to avoid solving.

Question 20 GRA-11 · Graphs of functions · spec MM8.7

The curves $y = 2^x$ and $y = x^2$ intersect at how many points?

Answer: A

They meet at $x = 2$ and $x = 4$ ($4 = 4$, $16 = 16$), and once for negative x (at $x = 0$: $1 > 0$; at $x = -1$: $\frac{1}{2} < 1$). For $x > 4$, 2^x grows faster. Three intersections.

Full method - how and why

Exponential versus quadratic. Test the obvious integer points and use end behaviour to be sure nothing is missed.

- 1 Try small integers: $x = 2$ gives $4 = 4$ ✓ and $x = 4$ gives $16 = 16$ ✓ - two intersections already.

why: $2^x = x^2$ has these two well-known solutions.

- 2 Check negative x : at $x = 0$, $2^0 = 1 > 0 = x^2$; at $x = -1$, $2^{-1} = \frac{1}{2} < 1 = x^2$. The order swaps, so there is a crossing between -1 and 0 .

why: A sign change in the difference $2^x - x^2$ guarantees a crossing.

- 3 Check beyond $x = 4$: 2^x grows faster than x^2 (e.g. at $x = 5$, $32 > 25$), so they never meet again.

why: Exponential growth eventually outpaces any polynomial.

- 4 Total: three intersections.

why: One negative, and $x = 2$, $x = 4$.

Where it goes wrong

- Finding only $x = 2$ and $x = 4$ and answering 2.
- Forgetting that x^2 is positive for negative x while 2^x stays small but positive.

Takeaway Test integer points, then check the two ends: a sign change in the difference means an extra crossing.